UNIVERSITÀ DI TORINO

chimica

Centro Interdipartimentale Amianti G. Scansetti
Università di Torino

TBM
TOXICITY and BIOCOMPATIBILITY of MATERIALS

Risonanze magnetiche

Reminder of Good Laboratory Practices

.....Some practical rules for better coexistence!

Workbenches – Sinks – Surfaces

- The workstation must be **thoroughly cleaned** before and after use. In the presence of colored reagents, it is recommended to place an appropriate layer of laboratory paper to prevent stains
- **Chemical reactions**, including mixtures (e.g., a beaker with reagents undergoing magnetic stirring), must **not** be **left unattended**. If the reaction requires extended time and the operator must leave, it is mandatory to leave a **written note** indicating the operator's **surname** and the **reagents** in the mixture
- Do not leave hot heating plates unattended. In this case, mark it with a paper note reading "**Caution: Hot Plate**"
- The **glassware** used by the operator should **be kept at the operator's station**
- **Never leave dirty glassware in the sink**
- The **sink sponge** should always be stored outside of the sink after use to prevent it from absorbing any chemicals
- **Windowsills** should always be kept clear! (especially when the window is open, but even when it is closed, it is a safety rule)

Balance room

- Both the **balance** and the **workbench** must be **thoroughly cleaned** (using the spray bottle). It is imperative that this area is kept clean. Laboratories use substances of various kinds, including those that are potentially or notoriously toxic and hazardous. Therefore, to ensure the safety and health of everyone, it is crucial to maintain a high standard of cleanliness in this area.
- **Turn off** the balance after use

Hoods

- In our laboratory, there are **two** fume hoods: one dedicated to the photocatalytic reactor, primarily used by the RM group, and another predominantly used by the TBM group. If one of the fume hoods needs to be used for an extended period, it is good practice to **coordinate in advance with other users** to ensure efficient use of the available resources.
- The **glass** of the fume hood must always be **fully lowered**
- The **workstation** under the fume hood must be **cleaned** before and after use
- Fume hoods should **not** be **used to store** reagents and samples
- If **reactions** are **ongoing**, write on a piece of paper: the operator's surname and the reagents used
- <u>At the end of the workday</u>, the fume hoods must be **turned off** using the main switch to avoid triggering the night alarm. Therefore, NO substances that require the operation of the fume hood should be left inside.

Commonly Used Equipment

- **Centrifuge**:

Use the weekly scheduling to make a reservation. The centrifuge is a shared piece of equipment, so it should only be reserved for the time strictly necessary for its use. If the equipment is not used, it is mandatory to cancel the reservation promptly. In any case, coordinating with surrounding users to optimize its use is good practice. Check at the end of the centrifugation that there has been no liquid leakage and, in case, clean thoroughly.

- **Ultrasonic baths**:

The water should always be clear and crystalline. Therefore, before use, if the water is dirty, it should be discarded, the container should be wiped with a paper towel, and fresh Elix water should be added.

Commonly Used Equipment

- **Ovens**:

- Use the ovens according to the instructions on the door (oven 1 for analytical glassware, and oven 2 for synthesis glassware and samples)

- The upper shelf of oven 1 is dedicated to EPR cells (and not other glassware).

- Remember to empty the oven in the morning

- **Automatic Pipettes**:

Pipettes are very useful instruments but require constant care and attention!

- Only use them after receiving proper **training**
- A set of pipettes is dedicated to **synthesis** (as indicated above), while the others are for "**cleaner" use**
- Before and after use, **ensure the tip ejector and cone is clean**. If in doubt, clean it with a piece of paper soaked in water. In any case, clean if potentially hazardous reagents have been used
- **Do not leave pipettes lying on the bench with tips full of liquid or dirty**. The pipette will easily become contaminated. In general, pipettes should never be left horizontally but should always be stored vertically

Miscellaneous

- **Waste**: Dispose of the waste from **organic solvents**, **halogenated organic solvents**, **acids**, **metals**, and **washing solutions** into the appropriate bins. The opening of the bins and the transfer of waste must be done UNDER THE FUME HOOD. Indicate the content of the waste in the waste book
- **Cancel the reservation**, if a reserved instrument (such as a centrifuge, EPR, muffle furnace, etc.) is no longer being used, to allow everyone to optimize their activities
- **Consumable stock:** For plasticware, glassware, gloves, reagents, etc., in general, for ALL products that are consumed and need to be reordered, indicate the remaining quantities on the list posted on the door of Lab 1 and **promptly notify the order coordinators of each group when stocks are running low**. Please note that, on average, it takes **more than one month** for common products (e.g., Falcon tubes) to arrive after an order is placed
- **Samples and personal solutions** kept on the bench or in the fridge must be clearly **labeled** with the operator's surname, preparation date, and content
- **Volumetric glassware should not be used for storing** solutions or samples, especially in the fridge. After preparing the solution, transfer it to a non-volumetric container (e.g., a bottle or Falcon tube)
- **Do not leave gloves or dirty materials lying around**

……And the last operator leaving the lab is kindly asked to turn off the lights!

Thank you for your cooperation!! ☺